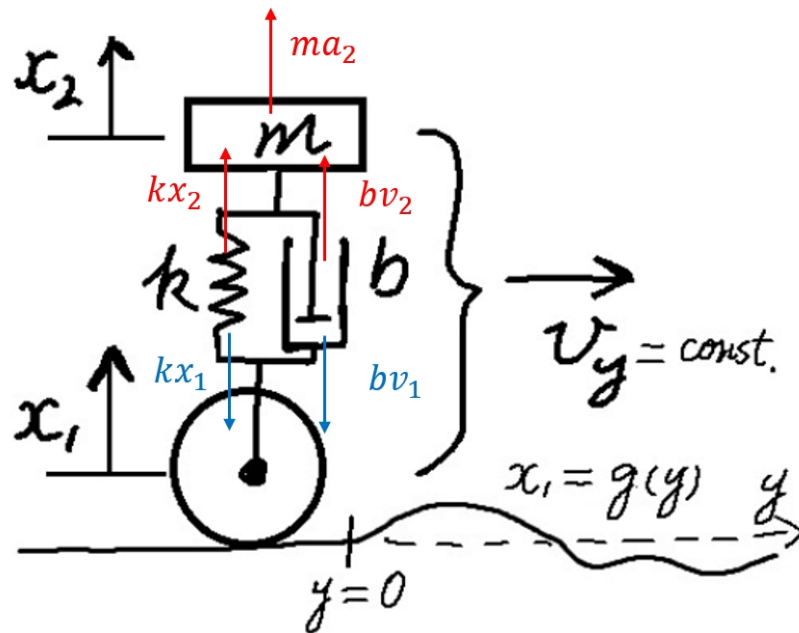


HW4 Discrete-time approximation of continuous-time systems

Part 1



According to the force diagram and laws of Newton's mechanics:

$$kx_1 + bv_1 = kx_2 + bv_2 + ma_2$$

Convert the formula to a differential equation:

$$kx_1 + b \frac{dx_1}{dx} = kx_2 + b \frac{dx_2}{dx} + m \frac{d^2 x_2}{dx^2}$$

or Laplace transformation:

$$kX_1(s) + bsX_1(s) = kX_2(s) + bsX_2(s) + ms^2 X_2(s)$$

Finally, we have

$$\frac{X_2(s)}{X_1(s)} = \frac{bs + k}{ms^2 + bs + k} = H(s)$$

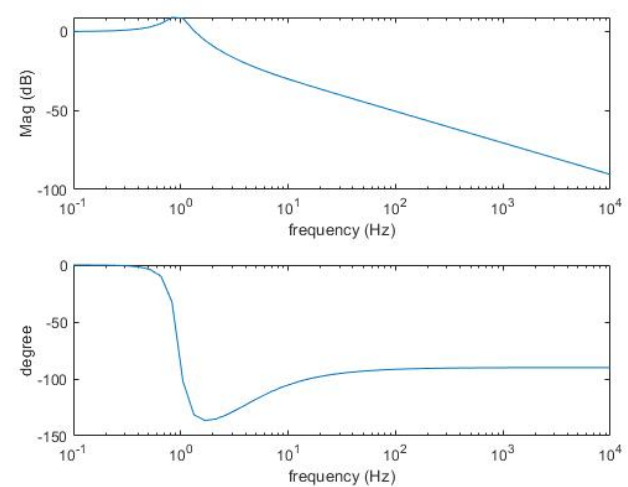
(b)

Draw the magnitude response and phase response of H.

```
f = logspace(-1,4,1e4);
s = 1j*2*pi*f;
H = (b*s + k)./(m*s.^2 + b*s + k);
mag = abs(H);
phs = angle(H);
% find peak
[~,ind] = max(mag);

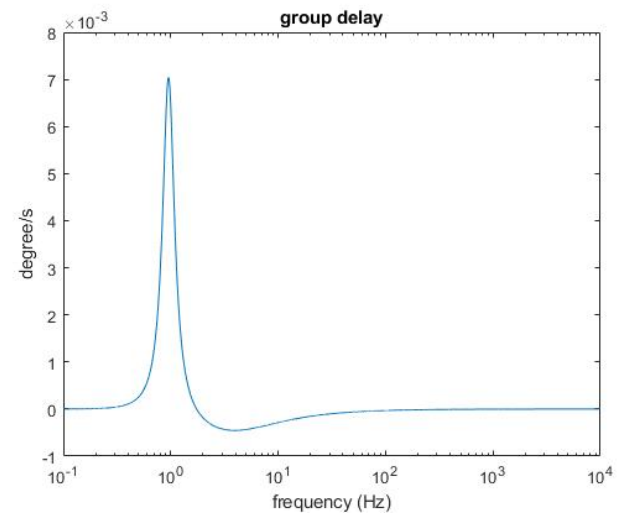
MaxValueAt = f(ind)
% MaxValueAt 0.9335 (Hz)
```

The maximal magnitude is at about 0.9335 Hz or 5.8654 rad/s. Since the order of denominator is greater than that of numerator, H(s) is approximately proportional to 1/s. Accordingly, as the frequency over 1 Hz, an increment of that degrades the magnitude of H(s). In contrast, magnitude of H(s) approaches to one or 0 dB when frequency is almost 0 Hz but gradually increases under 1 Hz. We can conclude that there exist a maximum such that the magnitude of H(s) achieves a peak then goes down.



(c)

According to the phase response, we can observe that phase roughly has a negative slope under 1.8 Hz but turn to increase over the frequency. Hence, by definition, group delay should similarly ascend before 1 Hz and gradually decline.



Part 2

(a)

$$kx_1 + b\frac{dx_1}{dx} = kx_2 + b\frac{dx_2}{dx} + m\frac{d^2x_2}{dx^2}$$

replace the first-order derivative term with $(x[n+1] - x[n-1])/2T$ and the second-order derivative term with $(x[n+1] - 2x[n] + x[n-1])/T^2$

$$kx_1[n] + b\frac{x_1[n+1] - x_1[n-1]}{2T} = kx_2[n] + b\frac{x_2[n+1] - x_2[n-1]}{2T} + m\frac{x_2[n+1] - 2x_2[n] + x_2[n-1]}{T^2}$$

$$Tbx_1[n+1] + 2T^2kx_1[n] - Tbx_1[n-1] = (Tb + 2m)x_2[n+1] + (2T^2k - 4m)x_2[n] + (2m - Tb)x_2[n-1]$$

$$x_2[n+1] = \frac{Tb}{Tb + 2m}x_1[n+1] + \frac{2T^2k}{Tb + 2m}x_1[n] - \frac{Tb}{Tb + 2m}x_1[n-1] - \frac{2T^2k - 4m}{Tb + 2m}x_2[n] - \frac{2m - Tb}{Tb + 2m}x_2[n-1]$$

or

$$x_2[n] = \frac{Tb}{Tb + 2m}x_1[n] + \frac{2T^2k}{Tb + 2m}x_1[n-1] - \frac{Tb}{Tb + 2m}x_1[n-2] - \frac{2T^2k - 4m}{Tb + 2m}x_2[n-1] - \frac{2m - Tb}{Tb + 2m}x_2[n-2]$$

(b)

Choose the sampling interval T with an appropriate length. Since a series of bumps that can be described as sine waves consisting of, the sampling interval can also be derived from the frequency of sine waves.

we combine $g(y)$ in different condition to a general representation

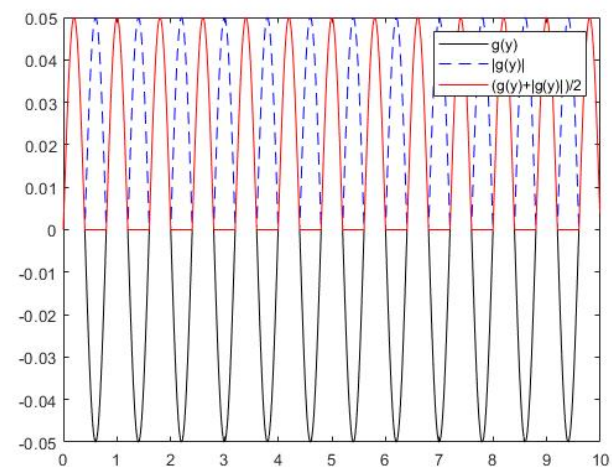
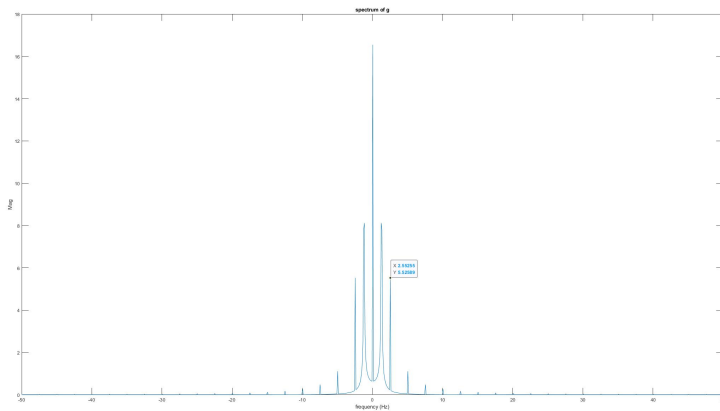
$$g(y) = D\sin(2\pi y/L) + |D\sin(2\pi y/L)| \text{ and } g(y+L) = g(y), \forall y > 0$$

The period of sine wave, $\sin(2\pi y/L)$, is about 0.8 s at speed 10 m/s, so there is a pair of delta pulses in frequency domain mathematically. As a result, by sampling theorem, sampling interval must be fewer than 0.4 s theoretically. However, not only the sinc function (due to finite signal seen in HW1) leading a side lobe extends bandwidth, but $g(y)$ additionally comprises an absolute term, $|\sin(2\pi y/L)|$, resulting in some harmonics. To obtain precise values, here we set sampling interval as 0.0001s which must suffice sampling theorem.

In practice, we can choose an absolutely short sampling interval in advance, drawing its spectrum to check the bandwidth, and then adjust sampling interval with reference to the Nyquist frequency.

For instance, after plotting, most information is included under 10 Hz, so we can further set sampling interval as 0.1 s. Although truncating some high frequency components would obtain an inaccurate wave, it reduces the requirement of hardware and still prevents from aliasing.

Draw the spectrum of $g(y)$



Note that $g(y)$ in the figure is defined as $g(y) = D\sin(2\pi y/L), 0 < y \leq L$ and $g(y+L) = g(y), \forall y > 0$.

Refer to problem (a)

$$x_2[n] = \frac{Tb}{Tb + 2m}x_1[n] + \frac{2T^2k}{Tb + 2m}x_1[n-1] - \frac{Tb}{Tb + 2m}x_1[n-2] - \frac{2T^2k - 4m}{Tb + 2m}x_2[n-1] - \frac{2m - Tb}{Tb + 2m}x_2[n-2]$$

Converted into Laplace form

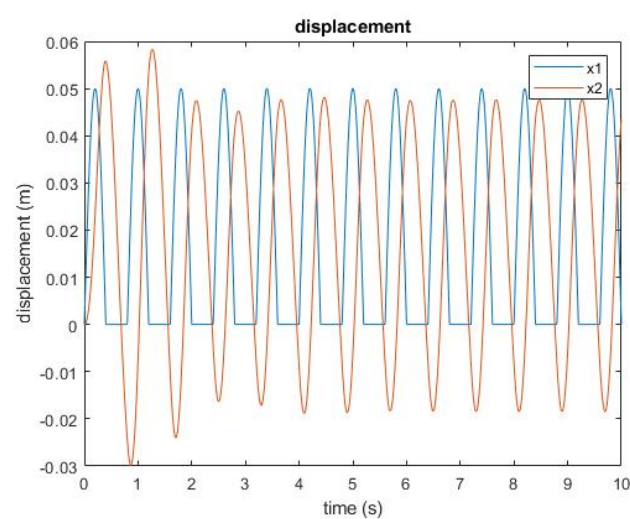
$$TbzX_1(z) + 2T^2kX_1(z) - Tbz^{-1}X_1(z) = (Tb + 2m)zX_2(z) + (2T^2k - 4m)X_2(z) + (2m - Tb)z^{-1}X_2(z)$$

$$\frac{X_2(z)}{X_1(z)} = \frac{Tbz + 2T^2k - Tbz^{-1}}{(Tb + 2m)z + (2T^2k - 4m) + (2m - Tb)z^{-1}}$$

```
% first way: calculate x2 using difference eq.
g = (D*sin(2*pi*y/L) + abs(D*sin(2*pi*y/L))) ./2;
x1 = g;
x2 = [0 0];
alpha = T*b + 2*m;
for ii = 3:length(g)
    x2(ii) = x1(ii)*T*b/alpha + x1(ii-1)*2*T^2*k/alpha - x1(ii-2)*T*b/alpha - x2(ii-1)*(2*T^2*k-4*m)/alpha - x2(ii-2)*(2*m-T*b)/alpha;
end
% another way: use built-in function filter
y = filter([T*b 2*T^2*k -T*b],[T*b+2*m 2*T^2*k-4*m 2*m-T*b],x1,[]);

MaxDisp = max(x2)
% MaxDisp = 0.0583 (m)
```

Finally, response of x2 is shown below



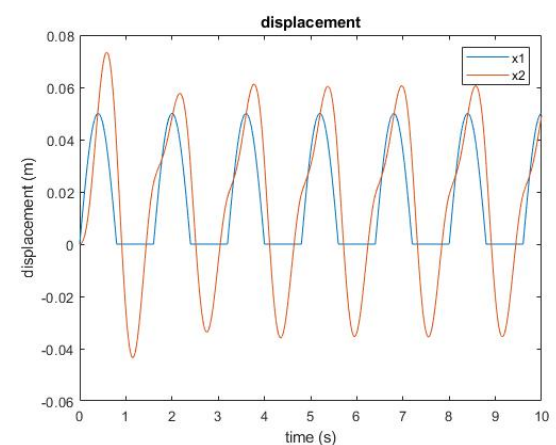
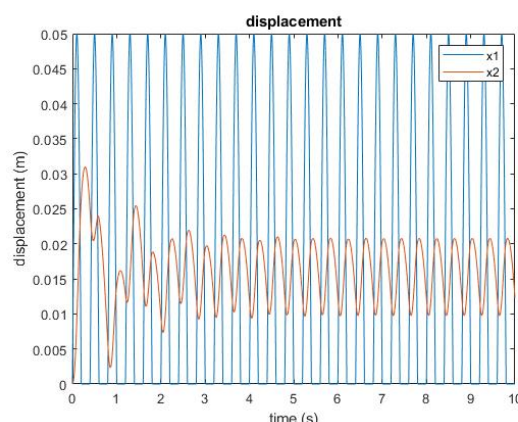
Maximal displacement is 5.83 cm.

(c)

The core concept is a car-suspension system takes the passenger comfort into account that can be viewed as a filter mitigating vibration. Input signal is vibration from bumps and output signal is the feedback acting on passengers. When the car is going at a constant speed 10 m/s, the maximal feedback is about 5.83 cm which is larger than the height of bumps.

Damp is commonly used to extenuate vibrations, and its performance is exactly proportional to speed. As the speed increases, damp has a larger force. In details, driving with high speed means **the displacement of x1 varies fast**, so does v1. Moreover, the feedback acting on passengers is transmitted from a spring rather than directly from bumps. A bump provides the perpendicular displacement to the connected spring, and then the compressed spring applies force on passengers in short. To reduce the force, a damp is connected with the spring and passengers. As per conversation, the feedback is awful when the speed isn't high enough that the damp can't supply a sufficient force against spring.

To justify our inference, we draw the responses of other cases. The left one describes the displacement response when v is set as 20 m/s and the other one is set as 5 m/s. As our expect, the right one has drastic vibration due to low speed and the damp has better performance when car runs fast.



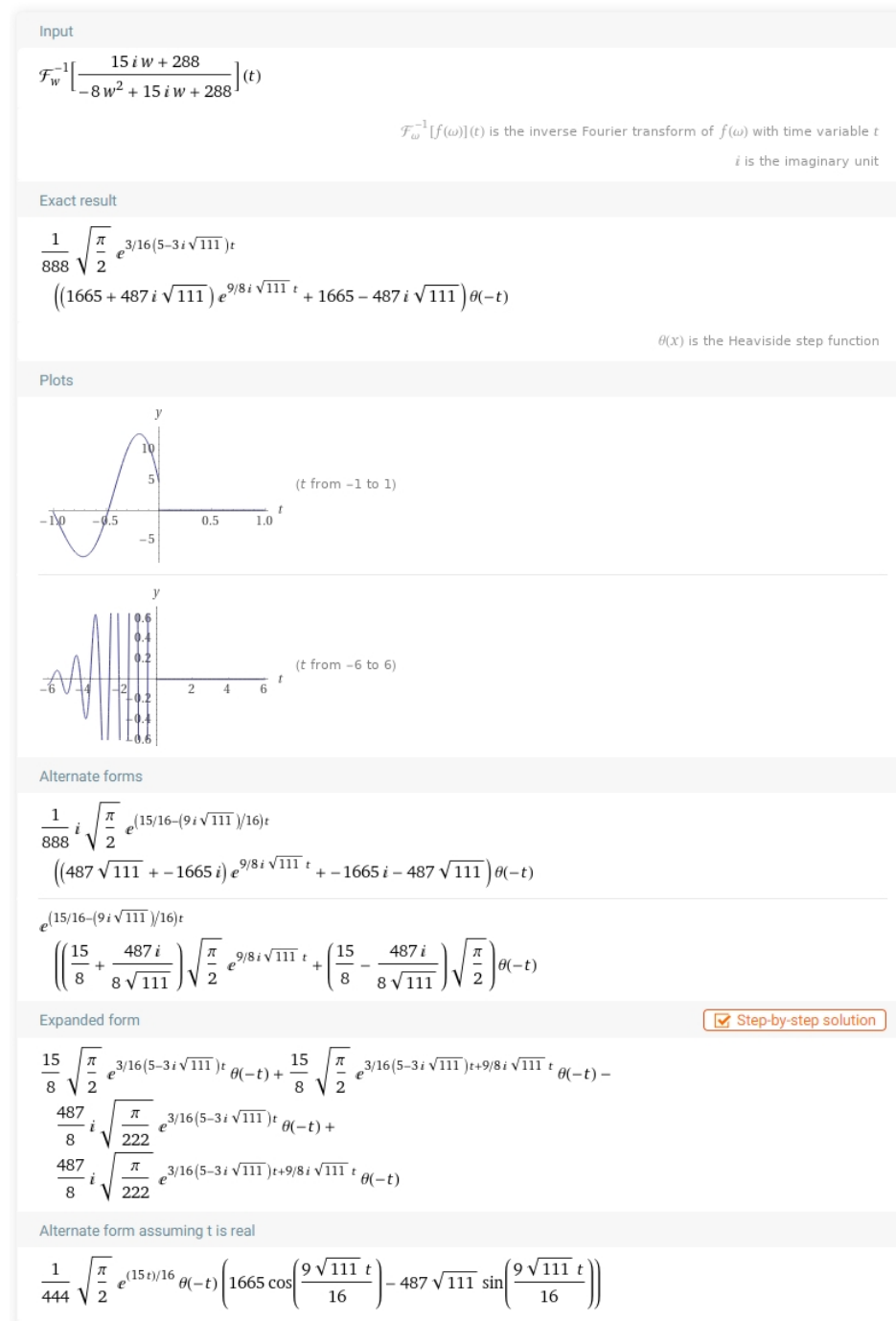
Part 3

(a)

first way : By online calculator or *ilaplace*:

$$\mathcal{L}^{-1}\left(\frac{15s + 288}{8s^2 + 15s + 288}\right) = \frac{(487\sqrt{111}\sin\frac{9\sqrt{111}t}{16} + 1665\cos\frac{9\sqrt{111}t}{16})e^{-\frac{15}{16}t}}{888}$$

another way: in part 1 , we replace s with $j2\pi f$, so inverse Fourier transform is available.



Note the result is

$$\frac{1}{444} \sqrt{\frac{\pi}{2}} e^{15t/16} \theta(-t) \left(1665 \cos\left(\frac{9\sqrt{111}t}{16}\right) - 487\sqrt{111} \sin\left(\frac{9\sqrt{111}t}{16}\right) \right)$$

where $\theta(t)$ is the unit step function $u(t)$, $-t$ can be replaced with t . Note that this calculator implement inverse Fourier transform using $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$ instead of $\frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$, so the above formula further needs to multiply a constant.

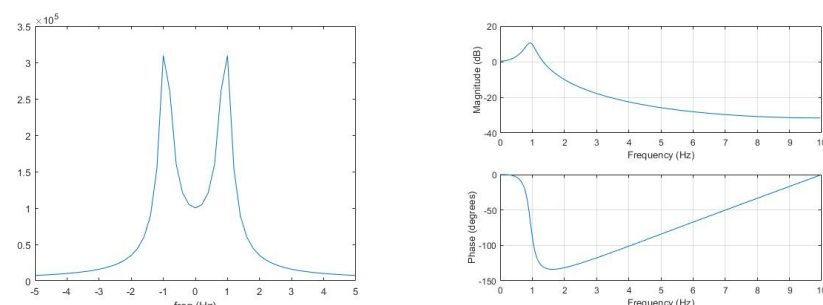
Finally, it is reformulated as

$$\frac{1}{888} e^{-15t/16} \left(1665 \cos\left(\frac{9\sqrt{111}t}{16}\right) + 487\sqrt{111} \sin\left(\frac{9\sqrt{111}t}{16}\right) \right) u(t)$$

It's equal to that derived from inverse Laplace transform.

(b)

Like problem (c) in part 2, now $h(t)$ consists of a sine wave at $9\sqrt{111}/16$ rad/s and exponential wave at $15/16$ rad/s, so we can know there is a delta pulse locating at 1.0924 Hz mathematically in frequency domain and further estimate sampling rate. Also owing to the effect of sinc, the Nyquist frequency must be larger than the theoretical value. To see the low frequency components, we set sampling frequency as 20 Hz.



Magnitude and phase response are plotted below and compared with that obtained in part 1. Because the phase response can't be taken out, we only verify it manually. Both are equivalent to that obtained in part 1.

Let's convert the unit of magnitude response into dB that would be more close to the bode plot in part 1.

At the beginning, we discuss this system in continuous time, so it can't be directly use *freqz* to draw its response. The default setting of frequency axis of *freqz* is from 0 to π , that means it can only analyze discrete-time signal. An alternative way is the built-in function *bode*, which is used to analyze continuous signals. In practice, most of received signals are required to an analog-to-digital converter. If we intend to perform *freqz* or processing in an application (software), sampling the continuous signal is necessary in advance. So in part 2, we convert a differential equation to a difference equation by forward difference and centered difference, equivalently, that is an analog-to-digital converter. Intrinsically, response of a continuous signal must be similar to that of corresponding discrete signal.

